Recurrent Thrifty Attention Network for Remote Sensing Scene Recognition

Liyong Fu, Dong Zhang, and Qiaolin Ye, *Member, IEEE*

***Abstract*— The self-attention mechanism has been empirically shown its effectiveness in a wide range of computer vision applications. However, it is usually criticized for the expensive computation cost. Although some revised methods are proposed in the recent past, they are not maturely applicable to remote sensing scene (RSS) images. To address this problem, in this article, we propose a simple yet effective context acquisition module, named thrifty attention, which can capture the long-range dependence efficiently and effectively. Moreover, a recurrent version for thrifty attention, termed recurrent thrifty attention (RTA), is further proposed to take the long-range multihop communications in space–time for RSS images. RTA is a general global contextual information acquisition module that can be used in any hierarchy of deep convolutional neural networks. To demonstrate its superiority, we deploy it to the classical ResNet and establish our proposed RTA Network (RTANet). Extensive experiments are carried out on two levels of the RSS recognition tasks, i.e., the image-level RSS classification and the instance-level RSS object detection. Compared with the standard self-attention mechanism, RTA can reduce at most 0.43 M model parameters while increasing a slight of model floating-point operations per second (FLOPs). Furthermore, results on RSS classification and object detection further verify the accuracy superiority of RTANet.**

***Index Terms*— Attention learning, convolutional neural networks (CNNs), object detection, remote sensing scene (RSS) classification, RSS recognition.**

I. Introduction

REMOTE sensing scene (RSS) recognition is one of the most fundamental research topics in the community of geographic image processing, which plays an irreplaceable role in various real-world applications, e.g., land management, forest assessment, and ecological monitoring [1], [2]. In recent years, Due to the tremendous progress in the domain of deep convolutional neural networks (CNNs), the performance for the vision-based image recognition applications is greatly improved [3]. For this reason, some specific RSS recognition tasks are also beneficiary, e.g., RSS classification [4], [5], object localization [6], and change detection [7], [8].

However, the standard CNNs have a latent imperfection in exploring the global contextual information, i.e., a single convolution module can only capture the short-range dependence at each hierarchy, which has been proved to be deficient for visual recognition [9]. Although this problem can be alleviated by superimposing multiple off-the-shelf convolution modules, the stacked structure may fetch disadvantages of the massive computation overhead and the optimization difficulty. Besides, to capture the long-range dependence (i.e., the global context) without increasing the computation cost, the dilated/atrous convolution [10] is proposed and successfully used in semantic segmentation [11]. However, the dilated/atrous convolution is essentially a subsampling-based operation, which may cause the problem of critical cues lost in practice [12].

To capture the long-range dependence by a single module, the specialized context acquisition modules are proposed, e.g., the nonlocal [13] and the self-attention mechanism [14]. The basic idea behind these two methods is almost the same. Specifically, as shown in Fig. 1(a), the pairwise affinities between every two feature positions (i.e., query and key) of the input image are first calculated by a similarity calculation approach. Then, a large-scale weighted context map is generated. After normalization, the context map is aggregated into each feature position (i.e., value) by a matrix multiplication. Although the above context acquisition methods have been proved to be effective in a wide range of computer vision tasks [15], [16], they are always criticized for their expensive computation cost [17]. For example, for an arbitrary input feature map $\mathbf{X}$ with the spatial size of $H \times W$ (H is the height and W is the width) and dimension of C, the computation complexities for the nonlocal and the self-attention mechanism in time and space are both $\mathcal{O}(CH^2W^2)$. Notably, this burden will be more serious when input images have a large scale.

To reduce the computation complexity of the specialized context acquisition module, downsampling is an effective approach. As in [18] and [19], feature maps are first sampled before processed by the attention module concretely, which corresponds to the criss-cross sampling and the pyramid sampling, respectively. Experimental results of semantic segmentation confirmed their effectiveness. Although the abovementioned two methods can reduce the computation

Manuscript received August 28, 2020; revised October 30, 2020; accepted November 29, 2020. Date of publication December 23, 2020; date of current version September 27, 2021. This work was supported in part by the Central Public-Interest Scientific Institution Basal Research Fund under Grant CAFYBB2019QD003, in part by the National Science Foundation of China under Grant 62072246 and Grant 61773210, and in part by the Qinglan and Six Talent Peaks Projects of Jiangsu Province. *(Liyong Fu and Dong Zhang contributed equally to this work.) (Corresponding author: Qiaolin Ye.)*

Liyong Fu is with the College of Information Science and Technology, Nanjing Forestry University, Nanjing 210037, China, and also with the Institute of Forest Resource Information Techniques, Chinese Academy of Forestry, Beijing 100091, China (e-mail: fuliyong840909@163.com).

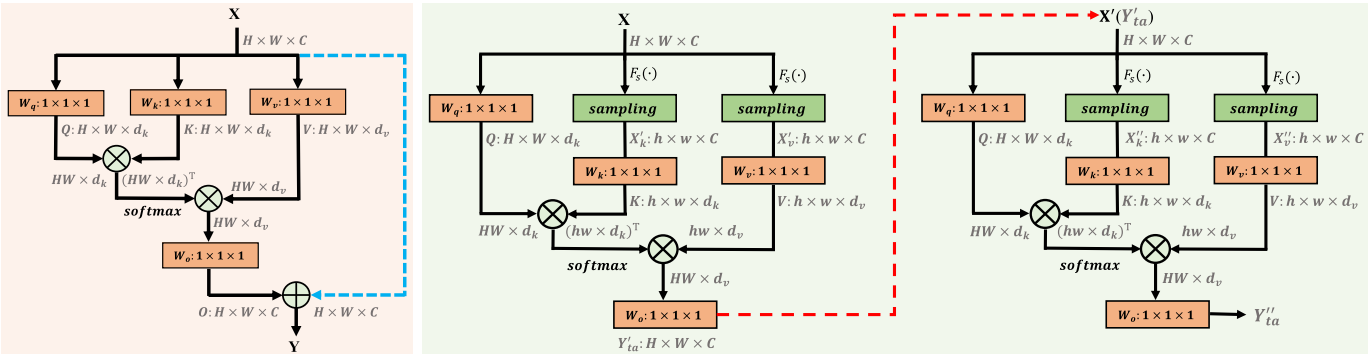
Dong Zhang is with the School of Computer Science and Engineering, Nanjing University of Science and Technology, Nanjing 210094, China (e-mail: dongzhang@njust.edu.cn).

Qiaolin Ye is with the School of Information Science and Technology, Nanjing Forestry University, Nanjing 210037, China (e-mail: yqlcom@njfu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2020.3042507

1558-0644 © 2020 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:23:37 UTC from IEEE Xplore. Restrictions apply.



(a) (b)

Fig. 1. (a) Illustrations of the standard self-attention module and (b) our proposed RTA module with loop $l = 2$, where $h \ll H$ and $w \ll W$. The blue dashed line denotes the residual connection, and the red dashed line denotes the data flow without any process. For the presentation simplicity, we skip over the residual connection in the RTA module, but this procedure actually exists after each loop. (a) Standard self-attention module. (b) Our proposed RTA module.

complexity within a certain range, they tolerate shortcomings of the extreme sparse features [18] or the complicated procedure [19] such that some detailed and key information may be lost. More importantly, due to the uncertainty and polytropism of objects in RSS images [1], the abovementioned methods are actually not maturely applicable to RSS recognition.

In this article, we propose a simple yet effective contextual information acquisition module, named thrifty attention, which is traced to the same origin as the nonlocal and the self-attention mechanism, but the context is aggregated efficiently and effectively. Specifically, as shown in the left part of Fig. 1(b), the main difference from the self-attention is that we first make a dynamic downsampling operation on the feature maps that will be embedded into key and value. In particular, the downsampling process is implemented by the off-the-shelf pooling function that comes from the PyTorch [20] deep learning platform. Then, we implement the same operation as the self-attention module. Moreover, to take the long-range multihop communications in space–time for input images, we further propose a recurrent version of thrifty attention, termed recurrent thrifty attention (RTA), which can easily capture the global long-range contextual information. As shown in the overall structure of Fig. 1(b), RTA is a general global context acquisition module that can be used in any hierarchy of deep CNNs.

To demonstrate the superiority of our proposed RTA, we implement it on the classical ResNet [3] and establish our proposed RTA network (RTANet) for RSS recognition. Experimental results on two levels of the RSS recognition tasks (i.e., the image-level RSS classification, and the instance-level RSS object detection) confirm that RTANet can improve the model performance significantly. Specifically, for RSS classification, RTANet-101 achieves the new state-of-the-art classification accuracy of 98.03%, 99.56%, 99.21%, 97.92%, and 93.61% on the test set of RSSCN7 [21], the UC Merced Land Use dataset (UCML) [22], WHU-RS19 [23], AID Data Set (AID) [24], and NWPU VHR-45 (NWPU-45) [25], respectively. As for RSS object detection, RTANet-101 achieves the mean average precision (mAP) over all categories of 0.7906 on the validation set of a Large-scale Dataset for Object Detection in Aerial Images (DOTA) [26] and 0.8420 on the test set of NWPU VHR-10 [27]. More importantly, in terms of the computational efficiency, compared with the standard self-attention mechanism, RTA can reduce at most 0.43 M model parameters, which is a very competitive result in the community of attention learning. The main contributions of this work are as follows.

1) We propose an efficient and effective RTA to capture the global long-range contextual information for RSS images.
2) We propose an end-to-end training RTANet. Experimental results demonstrate that RTANet can achieve a very competitive performance on a wide range of data sets of two levels of RSS recognition tasks.

II. Related Work

In this section, we briefly review two levels of the related tasks (i.e., the image level and the instance level) for RSS recognition (in Section II-A) as well as the attention-based learning approaches in the community of computer vision (in Section II-B).

A. RSS Recognition

RSS recognition aims to estimate the attribute information (e.g., the category, the quantity, and the location) of the ground objects in an input image via the feature understanding and analysis technologies [6]. In this article, we mainly discuss two levels of the RSS recognition tasks, i.e., the image-level RSS classification[1] and the instance-level RSS object detection. RSS classification is designed to predict a unique class label for each image. RSS object detection is an elaborate recognition task. It not only needs to recognize which objects are in the RSS image but also needs to locate where these objects are. Then, some bounding boxes are used to surround these objects. Compared to natural scenes, objects in RSS images have characteristics of the large uncertainty in the direction, the protean resolution in the size, and the incomplete cues in the appearance. Therefore, these factors lead to the RSS recognition an exceedingly challenging task.

The main challenge for RSS recognition is how to learn a robust feature representation with a strong discrimination [1].

[1]In this article, we only discuss the RSS classification problem of the single label, i.e., each image is set a label with only one category.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:23:37 UTC from IEEE Xplore. Restrictions apply.

From the features used point of view, we can mainly divide the existing RSS classification and RSS object detection methods into the following three categories: the shallow manual features, the deep learning features (mainly based on CNNs), and the combination of the abovementioned two categories. The shallow manual features mainly come from the appearance of input images, e.g., color, texture, and shape [7]. Moreover, some local manual features are also often used in RSS recognition, e.g., scale invariant feature transform (SIFT) [28], bag of features (BOF) [29], and vector of locally aggregated descriptors (VLAD) [30]. However, due to the limited representation ability of the shallow manual features, the model performance based on these features for RSS recognition is also stuck in a neck. To generalize the semantic information of images, methods based on the deep learning features are widely used in RSS recognition in recent years. Compared with the shallow manual features, deep learning features mainly have the advantages of the strong representation ability and rich semantic information inside. Therefore, the performance of RSS recognition is also improved in response [4]. In addition to using a single feature, the combination of the different features is also an effective approach. In particular, the combination of deep learning features and shallow manual features has attracted a lot of attention recently [31], [32]. Under this circumstance, shallow manual features are often regarded as an auxiliary to the deep learning features. Experimental results also confirm the effectiveness of this combination.

B. Attention Learning

Although CNNs have demonstrated its powerful learning ability in practice, the standard CNNs are limited by the local and short-range dependence information [33]. In recent years, researchers have found that capturing the long-range dependence is beneficial for visual recognition tasks. Squeeze-and-excitation networks [34] proposed a channel-wise attention mechanism, which used the global average pooling to calculate a weight for each channel of the feature map. Besides, spatial-wise attention is also a significant research topic. Vaswani *et al.* [14] used the self-attention mechanism (typical spatial-wise attention) in natural language translation and obtained a new state-of-the-art result. Moreover, Wang *et al.* [13] proposed a general nonlocal module to facilitate the performance of visual recognition. In addition to using an individual attention mechanism in CNNs, the combination of the channel- and spatial-wise attention is also a novel approach, such as dual attention network (DANet) [16] and discrimination oriented network (DONet) [35].

Although the self-attention mechanism is a useful approach in capturing long-range dependence, it is always criticized for its expensive computational cost. To address this problem, criss-cross attention network (CCNet) [18] and asymmetric nonlocal network [19] both adopted the idea of dimension reduction on feature maps and achieved satisfactory results on the semantic segmentation task. Our proposed thrifty attention can be traced to the same origin as the previous self-attention modules, but the contextual information is aggregated in a more efficient and effective style. Moreover, the proposed RTANet can be used to capture the global long-range contextual information.

III. Methodology

In this section, we will introduce details of our proposed RTANet for RSS recognition. In this section, we first make a brief review of the classical self-attention module on images (in Section III-A). We further propose the thrifty attention (in Section III-B), as its name suggests, which is a more efficient and effective method in acquiring the contextual information. After that, we present the RTA to capture the global long-range context of an input image (in Section III-C). Finally, we show the overall network architecture of the proposed RTANet in detail (in Section III-D).

A. Revisiting the Self-Attention Module

Suppose that an arbitrary RSS image is passed through a deep fully CNN and then is processed into a cluster of feature maps $\mathbf{X} \in \mathbb{R}^{H\times W\times C}$ as $\mathbf{X} = \{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_C\}$, where each feature map $\mathbf{x}_i$ is with the size of $H \times W$ and $i = 1, 2, \ldots, C$ is the feature dimension. As shown in Fig. 1(a), the self-attention first calculates a weighted context map, where each query position computes the pairwise affinities among all feature positions of the input image. After the normalization, this weighted context map is aggregated into the whole feature map by matrix multiplication. Then, the sum with the residual connection is used as its output. Therefore, the classical self-attention [14] (or the nonlocal [13]) is expressed as

$$\mathbf{O} = \underbrace{\text{Softmax}\big((\mathbf{X}\mathbf{W}_q)(\mathbf{X}\mathbf{W}_k)^T\big)}_{\mathbf{P}\in\mathbb{R}^{\text{HW}\times\text{HW}}}(\mathbf{X}\mathbf{W}_v) \tag{1}$$

$$\mathbf{Y} = \mathbf{O}\mathbf{W}_o + \mathbf{X} \tag{2}$$

where $\mathbf{W}_q, \mathbf{W}_k \in \mathbb{R}^{C\times d_k}$, and $\mathbf{W}_v \in \mathbb{R}^{C\times d_v}$ are three feature embedding functions, which are used to project the input feature maps $\mathbf{X}$ into three embedded feature spaces [14], i.e., *query*, *key*, and *value*, respectively. d_k and d_v are two hyperparameters, which are usually set to $C/2$. $\mathbf{P} \in \mathbb{R}^{HW\times HW}$ is the global context mapping matrix, in which each of its element denotes the similarity of two positions (pixels) in the input feature map $\mathbf{X}$. $\mathbf{W}_o \in \mathbb{R}^{d_v\times C}$ is a learnable parameter, which is used to increase the dimension of the feature map $\mathbf{O} \in \mathbb{R}^{H\times W\times d_v}$ whose features already contained the long-range contextual information into the same as $\mathbf{X}$. $\mathbf{Y} \in \mathbb{R}^{H\times W\times C}$ is the module output.

Although the standard self-attention can be used to capture long-range contextual information, the computational overhead is extremely expensive, which is intolerable in practice. To be specific, this overhead mainly comes from the two matrix multiplications, e.g., the computational complexities in time and space are both $\mathcal{O}(CH^2W^2)$. Especially for the problem of RSS recognition, the processed images are usually with the high resolution such that the computational complexity of the model is unacceptablely large. This is essentially not suitable in practice. To address this problem, in Sections III-B and C, we will propose a more efficient and effective RTANet in capturing the long-range contextual information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:23:37 UTC from IEEE Xplore. Restrictions apply.

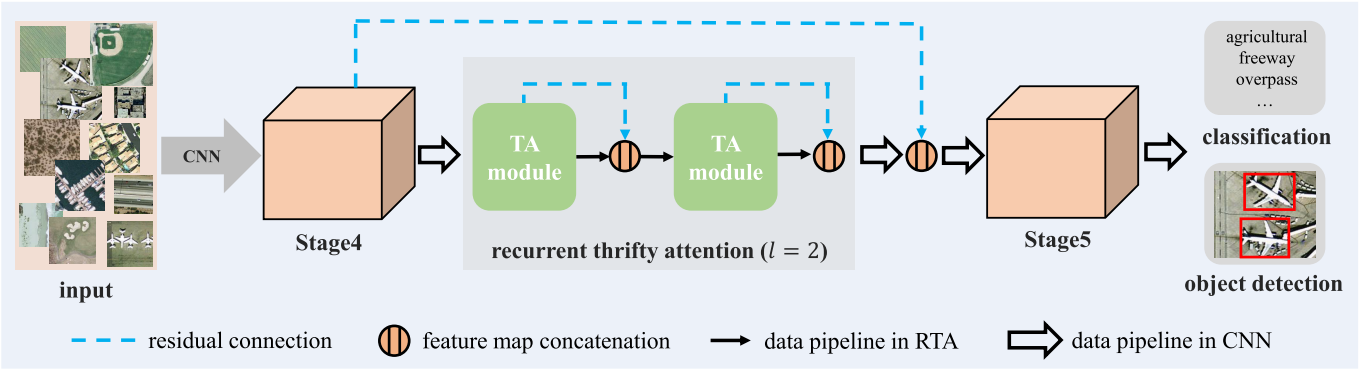


Fig. 2. Overview of our proposed RTANet. The input is a group of RGB RSS images and their corresponding annotations. After a deep feature extraction with a deep CNNs model, the proposed TAN (with the loop $l = 2$) is added to the output of the fourth stage with an additional residual connection. Finally, a specific head network is used for recognition on the output of the last stage. The whole network is trained in an end-to-end manner.

B. Thrifty Attention

As in [13] and [18], our proposed TA also considers the feature maps $\mathbf{X} \in \mathbb{R}^{H\times W\times C}$. As shown in the left part of Fig. 1(b), the main difference is that we first make a downsampling operation on the feature maps that will be embedded into *key* and *value*. Then, we perform the same operation as the self-attention. Therefore, the TA can be formulated as

$$\mathbf{X}'_k = F_s(\mathbf{X});\ \mathbf{X}'_v = F_s(\mathbf{X}) \tag{3}$$

$$\mathbf{O}'_{\text{ta}} = \underbrace{\text{Softmax}\big((\mathbf{X}\mathbf{W}_q)(\mathbf{X}'_k\mathbf{W}_k)^T\big)}_{\mathbf{P}'\in\mathbb{R}^{hw\times hw}}\big(\mathbf{X}'_v\mathbf{W}_v\big) \tag{4}$$

$$\mathbf{Y}'_{\text{ta}} = \text{cat}(\mathbf{O}'_{\text{ta}}\mathbf{W}_o, \mathbf{X}) \tag{5}$$

where $F_s(\cdot)$ denotes the sampling function, which is used to downsample spatial resolution of the input feature maps. $\mathbf{X}'_k$ and $\mathbf{X}'_v$ are two downsampled feature maps that correspond to the input feature maps on the *key* and *value* positions in the classical self-attention module, respectively. In this work, we use the standard pooling operation as the downsampling function. Besides downsampling the feature maps for *key* and *value*, in the feature fusion stage [i.e., the operation in (5)], we employ the feature concatenation instead of the simple summation. Existing literature works have proved that this substitution is beneficial to obtain a better fusion result [36]. $\mathbf{Y}'_{\text{ta}} \in \mathbb{R}^{H\times W\times C}$ is the output of the TA.

Assume that after the downsampling operation, the resolutions for the feature maps that will be embedded into *key* (i.e., $\mathbf{X}'_k$) and *value* (i.e., $\mathbf{X}'_v$) are both $h \times w \times C$, where $h \ll H$ and $w \ll W$. Then, the computational complexities in time and space of the TA are reduced from $\mathcal{O}(CH^2W^2)$ into $\mathcal{O}(Ch^2w^2)$ and the spatial sampling rate $r = h/H = w/W$. TA is a general context acquisition module that can be used in any hierarchy of a CNNs model. Moreover, compared with the standard self-attention, TA is more efficient and effective and with less computational overhead and GPU memory usage.

C. Recurrent Thrifty Attention

Although the TA can capture the contextual information successfully, as discussed in [13], it is difficult for the distant positions of the local model (e.g., the self-attention and the TA) to take the long-range multihop communications in space–time, which are proved to be beneficial for visual recognition [9], [15]. To achieve this goal, we further present an RTA, which is based on the proposed TA as mentioned earlier.

The main idea behind the RTA is that we repeat the TA multiple times, where the input of the latter loop is the output of the previous one. Fig. 1(b) shows a two-loop ($l = 2$) RTA module. For the convenient presentation, we skip over the residual connection, but this procedure actually exists. The input of the first loop is $\mathbf{X}$. After the same operation as the TA, it then outputs $\mathbf{Y}'_{\text{ta}}$. In the second loop [i.e., the right part of 1 (b)], the input is $\mathbf{Y}'_{\text{ta}}$ ($\mathbf{X}'$) and outputs $\mathbf{Y}''_{\text{ta}}$ after the same operation as the first loop. The second loop introduced above can be expressed as

$$\mathbf{X}''_k = F_s(\mathbf{Y}'_{\text{ta}});\ \mathbf{X}''_v = F_s(\mathbf{Y}'_{\text{ta}}) \tag{6}$$

$$\mathbf{O}''_{\text{ta}} = \underbrace{\text{Softmax}\big((\mathbf{X}\mathbf{W}_q)(\mathbf{X}''_k\mathbf{W}_k)^T\big)}_{\mathbf{P}''\in\mathbb{R}^{hw\times hw}}\big(\mathbf{X}''_v\mathbf{W}_v\big) \tag{7}$$

$$\mathbf{Y}''_{\text{ta}} = \text{cat}(\mathbf{O}''_{\text{ta}}\mathbf{W}_o, \mathbf{X}). \tag{8}$$

Hereby, we should note that the residual connection is actually added after each loop. The reason is that the existing literature works have proved that this repetitive procedure (i.e., adding a residual connection after each loop) is beneficial to the model optimization and beneficial for the visual recognition [37], [38]. In addition, the RTA is not limited to two loops, and the number of loops can be adjusted according to the specific needs.

As a preferable version of the thrifty attention, RTA can easily capture the global long-range contextual information and leads to better performance than the expensive self-attention mechanism but with less computational cost (details can be found in Section IV). In addition, the RTA module has the characteristic of plug-and-play, and therefore, it can be optimized in an end-to-end manner within any CNNs.

D. Network Architecture

Fig. 2 shows the network architecture of our proposed RTANet for RSS recognition. The input is a group of RGB RSS images and their corresponding annotations (e.g., the image classes or object bounding boxes). The backbone network is the classical ResNet [3], whose effectiveness has been verified in a large number of computer vision tasks. Although our proposed RTA can be inserted into any hierarchy

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:23:37 UTC from IEEE Xplore. Restrictions apply.

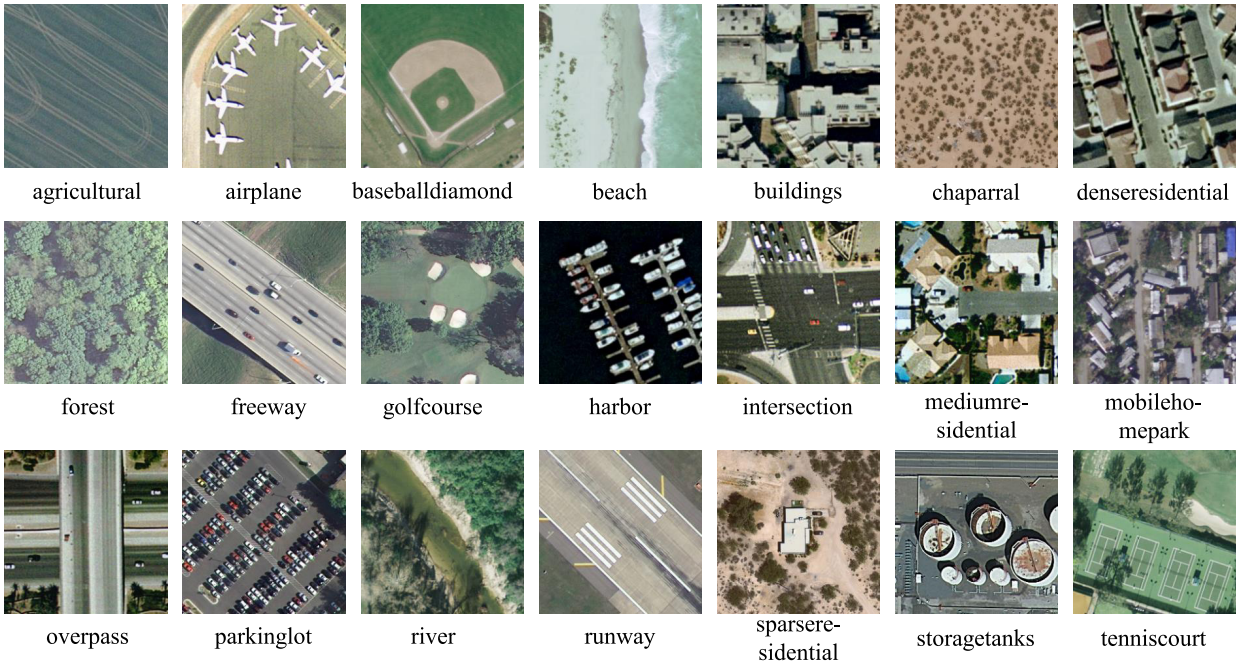


Fig. 3. Some samples of the experimental data set for RSS classification. Images are from the UCML data set.

of a CNNs model, its influence on the model performance is actually different at different hierarchies. As suggested in [13] and [18], it is an effective choice to add the attention-based module to the output feature maps of the last convolutional layer in the fourth stage[2] (more details on this point can be found in Section IV-A.6), whose feature size is 1/16 of the input. Therefore, we add the RTA into the output of the fourth stage in the classical ResNet network and then produce the proposed RTANet. Moreover, as in [40], we add an additional residual connection from the fourth stage to the output of RTA. This operation is beneficial to the model optimization [38]. Finally, based on the backbone network, we add a specific head network (i.e., RSS classification and RSS object detection in this work) on the output of the last stage for recognition. The source code for our model will be publicly available after the publication.

IV. Experiments

To demonstrate the efficiency and effectiveness of our proposed RTANet, we implement it on two levels of the RSS recognition tasks, i.e., the image-level RSS classification (in Section IV-A) and the instance-level RSS object detection (in Section IV-B).

A. Experiments on Image-Level RSS Classification

1) Data Sets: Five RSS classification data sets are selected as the experimental data. Specifically, RSSCN7 [21] contains a total of 2800 images of seven categories, each of which has the resolution of 400 × 400. UCML [22] contains a total of 2100 images of 21 categories, each of which has the resolution of 256 × 256. Some samples of this data set are shown in Fig. 3. WHU-RS19 [23] contains a total of 950 images in 19 categories, each of which has the resolution of 600 × 600. AID [24] contains a total of 10 000 images of 30 categories, each of which has 600 × 600 pixels. NWPU-RESISC45 (NWPU-45) [25] contains a total of 45 categories in 31 500 images, each of which has the resolution of 256 × 256. For each data set, we randomly divide it into 80% for training, 10% for validation, and 10% for testing. As in [1], the data augmentation is performed in the training set, i.e., flip (horizontally, vertically, and diagonally of 45° and 135°) and rotation (45°, 90°, 135°, 180°, 225°, and 270°).

2) Evaluation Metrics: To avoid the randomness of the experimental results, we run each experiment five times and report the mean classification accuracy (Acc) and standard deviation (STD) for the primary evaluation metrics. Results on the validation and test are expressed as ValAcc and TestAcc, respectively. Besides, to verify the computational complexity, we also make comparisons on the model parameters (params) and the floating-point operations per second (FLOPs).

3) Backbone Networks: The classical ResNet-50 [3] is selected as the primary backbone in the ablation study. Besides, to compare with the state-of-the-art methods, we also choose ResNet-101 as a backbone network, which can also verify the effectiveness of RTA in deeper backbones. In our experiments, all backbone networks are pretrained on ImageNet [41].

4) Baselines: In RSS classification, we set two baselines. Baseline-1 is the classical ResNet [3]. Baseline-2 is the nonlocal network [13] with ResNet as the backbone, which plays the role as the self-attention mechanism [14]. Except that the experimental data set is changed, the experimental settings of the baseline methods are consistent with their articles.

[2]We follow feature pyramid networks (FPN) [39] to define that layers generating feature maps with the same resolution are in the same network stage.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:23:37 UTC from IEEE Xplore. Restrictions apply.

5) Implementation Details: All experiments are carried out on the PyTorch [20] deep learning platform. All networks are trained in an end-to-end manner using four NVIDIA Tesla P100 GPUs with the Synchronized Batch Normalization (SyncBN) [40]. As in [4], to make the network converge quickly, all the experimental networks are first pretrained on the ImageNet [41] and then fine-turned with a specific RSS data set with the input image size of 256×256. In optimization, to enhance the stability of the deep learning model, the warm-up heuristic [42] is used in the training procedure. Specifically, we first set a small learning rate (0.001 in this article) at the first five training iterations. Then, we adopt the Nesterov optimizer with momentum $= 0.9$ and the initial learning rate is set to 0.01 and the rate decay is set to 0.001 in every 10 000 iterations. Besides, we also set the learnable parameter γ into 0 in the batch normalization for scale transformation, which is usually used after each residual block of the ResNet. The batch size is set to 64. The weight decay is 1e-4 and the maximum training iteration is set to 80 000.

6) Ablation Study: To explore the influence of different components on the experimental results and find out their optimal conditions, we implement the ablation study on RSSCN7, which aims to 1) select the most suitable downsampling approach for RTANet (including the mix one); 2) find out the applicable downsampling rate r that matches the RSS images; 3) find out the reasonable loop times (considering both the model efficiency and accuracy); 3) confirm which stage is suitable for adding RTA module; and 4) verify the effectiveness of the whole RTA module.

a) Selection of the sampling method: In the first experiment, we set the spatial sampling rate $r = 0.5$. As in [13], only one thrifty attention module is added into the output of the fourth stage. In addition to three common pooling functions (i.e., the max pooling, the average pooling, and the median pooling), in this article, we introduce a combined pooling approach named mix pooling. Mix pooling is averaged from the abovementioned three pooling functions. Specifically, the input images are first downsampled through the three pooling functions and then use them as the output of the mix pooling after averaging.

Table I shows the effect of different pooling functions on the experimental performance. As we can observe that: 1) compared to the baseline-1, all models with an attention mechanism are beneficial for improving the performance; 2) compared to the baseline-2, the downsampling operation of feature maps can not only further improve the model performance but also reduce the computational cost; and 3) among these pooling functions, the introduced mix pooling has the maximal performance improvement (i.e., 1.08% ValAcc and 0.96% TestAcc higher than the baseline-2) with a gleaming computational cost (i.e., 1.97 M params and 0.2 G FLOPs cheaper than the baseline-2). When compared to the three common pooling functions, mix pooling has the common advantages of the other three pooling functions, which will not cause the loss of the crucial cues nor have a complicated calculation procedure. Therefore, the proposed TA module is very suitable for RSS images that have the uncertainty and polytropic objects. Considering the important model performance, in the following experiments, we use mix pooling as the downsampling function.

TABLE I

EFFICACY OF DIFFERENT DOWNSAMPLING METHODS. THE BACKBONE IS RESNET-50

methods	ValAcc±STD (%)	TestAcc±STD (%)	FLOPs	params
baseline-1	93.87 ± 0.34	93.23 ± 0.25	**8.2** G	**25.65** M
baseline-2	95.16 ± 0.21	94.87 ± 0.32	8.9 G	30.15 M
max pooling	95.49 ± 0.33	94.96 ± 0.26	8.5 G	27.41 M
average pooling	95.65 ± 0.16	95.05 ± 0.34	8.5 G	27.41 M
median pooling	95.59 ± 0.57	95.10 ± 0.26	8.5 G	27.41 M
mix pooling	**96.24 ± 0.20**	**95.83 ± 0.38**	8.7 G	28.18 M

TABLE II

EFFECT OF THE DOWNSAMPLING RATE r ON THE MODEL PERFORMANCE. THE BACKBONE IS RESNET-50

settings	ValAcc±STD (%)	TestAcc±STD (%)	FLOPs	params
baseline-1	93.87 ± 0.34	93.23 ± 0.25	**8.2** G	**25.65** M
baseline-2	95.16 ± 0.21	94.87 ± 0.32	8.9 G	30.15 M
$r = 0.5$	**96.24 ± 0.20**	**95.83 ± 0.38**	8.7 G	28.18 M
$r = 0.25$	95.26 ± 0.24	94.57 ± 0.72	8.5 G	27.53 M
$r = 0.125$	94.17 ± 0.35	93.56 ± 0.25	8.3 G	26.19 M

b) Efficacy of the sampling rate: After the input images (with the input size of 256×256) are downsampled 16 times in Stage4, the output feature map size is 16×16. It is obvious that the downsampling rate of the feature maps will influence the model performance. Hence, in this section, we will explore the effect of the downsampling rate r on the model performance as well as the computational cost. Here, we should note that the baseline-2 corresponds to the setting of $r = 1$, i.e., there are no feature maps that are downsampled. Experimental results are shown in Table II. $r = 0.5$ achieves the best performance, i.e., 96.24% ValAcc and 95.83% TestAcc. Besides, we can observe that as the sampling rate continues to decline on the base of 0.5, although the computational cost decreases, the performance also decreases in response, i.e., 95.26% ValAcc and 94.57% TestAcc when $r = 0.25$ and 94.17% ValAcc and 93.56% TestAcc when $r = 0.125$. One possible explanation is that there are not enough instance cues under these cases. Therefore, we set the downsampling rate $r = 0.5$ in the following experiments.

c) Which stage to add TA?: Although in [13], the author concluded that adding the attention module to Stage4 can achieve the best performance, it is based on the natural scenes that have some differences from RSS images. Therefore, in this section, we will verify the correctness of this conclusion on RSS images. The TA module is added on different stages of the ResNet, i.e., the backbone network. In particular, Stage1 is not considered because the resolution of the feature maps for this stage is very high, which will cause a large amount of computational overhead. Therefore, the attention module is generally not added to this stage. Results of adding the attention module to Stage2–Stage5 are shown in Table III. We can observe that although the performance in Stage2 and Stage3 is better than that in Stage4, the computational overhead is very expensive.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:23:37 UTC from IEEE Xplore. Restrictions apply.

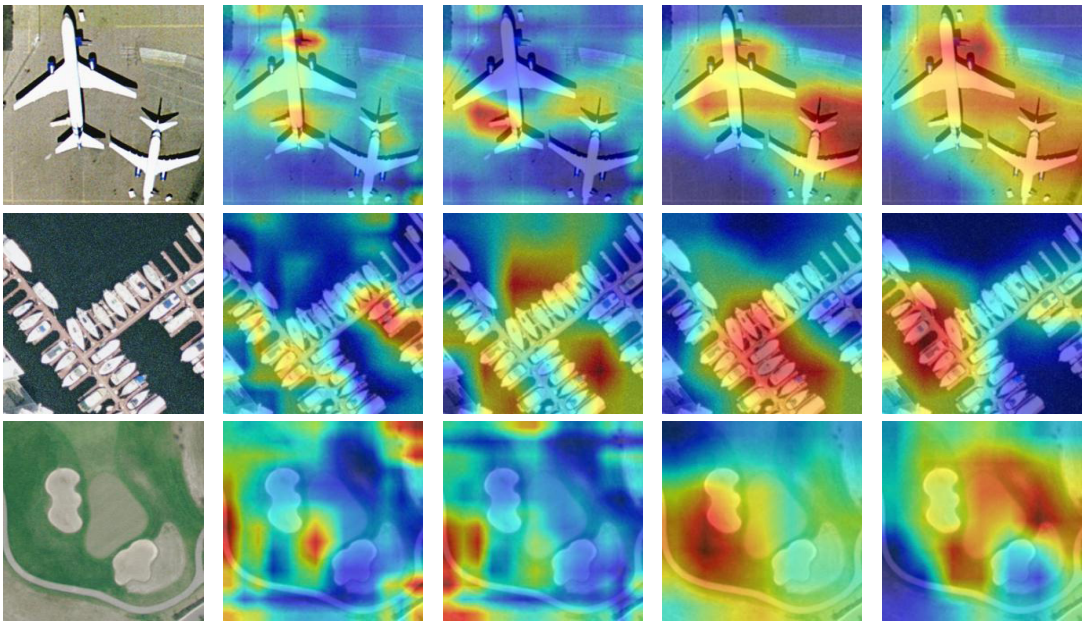


Fig. 4. Gradient-weighted class activation mappings of baselines and our proposed RTANet for RSS classification. Images are from the UCML data set. The reddish areas mean that these areas play a significant role in image classification and vice versa. (a) Input. (b) Baseline-1. (c) Baseline-2. (d) RTANet (l = 1). (e) RTANet (l = 2).

TABLE III
PERFORMANCE ON DIFFERENT STAGES. THE BACKBONE IS RESNET-50

settings	ValAcc±STD (%)	TestAcc±STD (%)	FLOPs	params
baseline-1	93.87 ± 0.34	93.23 ± 0.25	**8.2** G	**25.65** M
baseline-2	95.16 ± 0.21	94.87 ± 0.32	8.9 G	30.15 M
Stage2	**97.04 ± 0.37**	**96.83 ± 0.26**	9.1 G	33.62 M
Stage3	96.72 ± 0.40	96.28 ± 0.28	8.9 G	31.35 M
Stage4	96.24 ± 0.20	95.83 ± 0.38	8.7 G	28.18 M
Stage5	95.74 ± 0.25	95.17 ± 0.36	8.6 G	28.06 M

TABLE IV
EFFICACY OF THE RE CURRENT LOOP l IN RTA. THE BACKBONE IS RESNET-50

settings	ValAcc±STD (%)	TestAcc±STD (%)	FLOPs	params
baseline-1	93.87 ± 0.34	93.23 ± 0.25	**8.2** G	**25.65** M
baseline-2	95.16 ± 0.21	94.87 ± 0.32	8.9 G	30.15 M
$l = 1$	96.24 ± 0.20	95.83 ± 0.38	8.7 G	28.18 M
$l = 2$	97.17 ± 0.37	97.03 ± 0.42	8.9 G	29.72 M
$l = 3$	97.62 ± 0.36	97.36 ± 0.34	9.1 G	30.91 M
$l = 4$	**97.93 ± 0.35**	**97.60 ± 0.21**	9.2 G	31.67 M

Therefore, to balance the computational complexity and the model performance, adding the TA module to Stage4 is an idealized cost-effective solution, which is consistent with the conclusion in [13]. Besides, from Table III, we can observe that the model performance in Stage5 is lower than the baseline. One possible explanation is that the resolution of feature maps in Stage5 is too low, which may cause some instances that cannot provide enough cues [13].

d) Efficacy of the loop: The above three ablation studies determine a basic TA module. In this section, we explore the efficacy of the recurrent loop l in the RTA module, which is added to Stage4 of the backbone network. Table IV shows the model performance as well as the increment of FLOPs and params. We can observe that adding one TA module (i.e., when $l = 1$) into the baseline-1 can increase 1.08% ValAcc and 0.96% TestAcc with cheaper computational cost (the decrement of 1.97 M params and 0.2 G FLOPs) than beseline-2. Furthermore, after adding one more loop (i.e., when $l = 2$), the model can obtain the performance gain of 2.01% ValAcc and 2.16% TestAcc. When compared to baseline-2, the model params decrease by 0.43 M and the FLOPs are changeless. Hereafter, as the number of the recurrent loop increases, the obtained performance gain will gradually decrease. This phenomenon is consistent with the one in [18]. Compared with the baseline methods, the abovementioned experimental results confirm the effectiveness and efficiency of our proposed RTA in RSS recognition quantificationally.

Fig. 4 shows the gradient-weighted class activation maps for baselines and our proposed RTANet in RSS classification on the UCML data set. The scarlet areas denote that these areas play a significant role in image classification and vice versa. From the visualization results, we can observe that RTANet can activate more discriminative areas when compared to these two baselines, which indicating that more cues of the object are used in RTANet classification. Furthermore, compared with the results of $l = 1$, the results of $l = 2$ also show that more areas are activated. Therefore, the effectiveness of our proposed RTANet can be verified qualitatively. In the following experiments, to balance the performance and the computational complexity, we will set $l = 2$.

e) Effectiveness of RTA: In the training stage, three tricks (i.e., warm-up [42], SyncBN [40], and zero γ) are used to make the model converge quickly or stable. In this section, we will report an ablation study on these three tricks as

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:23:37 UTC from IEEE Xplore. Restrictions apply.

TABLE V
EFFICACY OF STAGE. "SYNCBN" DENOTES THE SYNCBN. THE BACKBONE IS RESNET-50

methods	warm-up [42]	SyncBN [40]	zero γ	ValAcc $\pm$ STD (%)	TestAcc $\pm$ STD (%)
baseline-1	✗	✗	✗	93.87 $\pm$ 0.34	93.23 $\pm$ 0.25
RTANet-50	✗	✗	✗	96.26 $\pm$ 0.31	95.82 $\pm$ 0.41
RTANet-50	✓	✗	✗	96.72 $\pm$ 0.29	96.36 $\pm$ 0.19
RTANet-50	✗	✓	✗	97.02 $\pm$ 0.37	96.67 $\pm$ 0.36
RTANet-50	✗	✗	✓	96.81 $\pm$ 0.20	96.55 $\pm$ 0.51
RTANet-50	✓	✓	✗	97.30 $\pm$ 0.29	97.02 $\pm$ 0.25
RTANet-50	✓	✗	✓	97.17 $\pm$ 0.21	96.89 $\pm$ 0.33
RTANet-50	✗	✓	✓	97.52 $\pm$ 0.31	97.27 $\pm$ 0.19
RTANet-50	✓	✓	✓	**97.93 $\pm$ 0.35**	**97.60 $\pm$ 0.21**

TABLE VI
COMPARISON TO STATE-OF-THE-ART METHODS ON THE TEST SET. "†" MEANS THAT THIS IS OUR REIMPLEMENTED RESULT

Methods	RSSCN7 [21]	UCML [22]	WHU-RS19 [23]	AID [24]	NWPU-45 [25]
baseline-1 [3]	93.23 $\pm$ 0.25	96.62 $\pm$ 0.26	97.17 $\pm$ 0.17	92.26 $\pm$ 0.19	90.50 $\pm$ 0.08
baseline-2 [13]	94.87 $\pm$ 0.32	98.01 $\pm$ 0.23	98.67 $\pm$ 0.53	93.82 $\pm$ 0.24	91.91 $\pm$ 0.16
CoordConv† [43]	94.11 $\pm$ 0.23	97.37 $\pm$ 0.33	98.24 $\pm$ 0.17	93.27 $\pm$ 0.35	91.17 $\pm$ 0.06
2D Sine† [44]	94.07 $\pm$ 0.15	97.31 $\pm$ 0.26	97.92 $\pm$ 0.25	93.01 $\pm$ 0.20	91.15 $\pm$ 0.36
GCNet† [15]	94.97 $\pm$ 0.13	98.25 $\pm$ 0.19	98.90 $\pm$ 0.15	93.91 $\pm$ 0.22	92.23 $\pm$ 0.30
Multi-head† [9]	95.37 $\pm$ 0.19	98.79 $\pm$ 0.21	98.91 $\pm$ 0.20	94.67 $\pm$ 0.23	92.33 $\pm$ 0.30
PANet† [1]	96.17 $\pm$ 0.04	99.21 $\pm$ 0.18	98.91 $\pm$ 0.17	97.05 $\pm$ 0.30	92.61 $\pm$ 0.25
CCNet† [18]	95.89 $\pm$ 0.27	99.31 $\pm$ 0.40	98.95 $\pm$ 0.16	97.41 $\pm$ 0.24	92.96 $\pm$ 0.40
AsymNet† [19]	96.36 $\pm$ 0.33	99.42 $\pm$ 0.19	98.96 $\pm$ 0.30	97.52 $\pm$ 0.31	92.92 $\pm$ 0.32
Discriminative CNNs [45]	93.98 $\pm$ 0.14	98.93 $\pm$ 0.17	98.51 $\pm$ 0.66	96.89 $\pm$ 1.20	91.89 $\pm$ 0.31
CNN + Deep Forest [31]	94.25 $\pm$ 0.19	98.97 $\pm$ 0.15	98.62 $\pm$ 0.30	95.67 $\pm$ 0.36	89.66 $\pm$ 0.21
Sifting GAN [46]	92.82 $\pm$ 0.17	97.85 $\pm$ 0.16	98.78 $\pm$ 0.26	90.64 $\pm$ 0.47	90.01 $\pm$ 0.21
VGG-CapsNet [47]	92.03 $\pm$ 0.25	98.81 $\pm$ 0.22	97.02 $\pm$ 0.20	91.63 $\pm$ 0.19	89.18 $\pm$ 0.14
BAFF [48]	91.73 $\pm$ 0.15	95.48 $\pm$ 0.22	98.69 $\pm$ 0.25	93.56 $\pm$ 0.31	89.66 $\pm$ 0.33
Two-stream Fusion [49]	93.14 $\pm$ 0.17	98.02 $\pm$ 1.03	98.79 $\pm$ 0.99	94.58 $\pm$ 0.25	83.02 $\pm$ 0.14
GCFs + LOFs [32]	95.59 $\pm$ 0.49	99.00 $\pm$ 0.35	97.89 $\pm$ 0.24	92.48 $\pm$ 0.38	90.64 $\pm$ 0.34
RTANet-50 (**ours**)	97.60 $\pm$ 0.21	99.49 $\pm$ 0.32	99.04 $\pm$ 0.25	97.89 $\pm$ 0.20	93.05 $\pm$ 0.15
RTANet-101 (**ours**)	**98.03 $\pm$ 0.27**	**99.56 $\pm$ 0.26**	**99.21 $\pm$ 0.19**	**97.92 $\pm$ 0.19**	**93.61 $\pm$ 0.36**

well as their arbitrary combinations. Experimental results are shown in Table V. We can observe that every trick contributes to the performance improvement, where the SyncBN gives the maximum improvement (i.e., 3.15% ValAcc and 3.44% TestAcc). Besides, any combination of them achieves the better performance than any individual one. Not surprisingly, the combination of the warm-up and the SyncBN and the zero γ achieves the best performance (4.06% ValAcc and 4.37% TestAcc).

7) Comparisons With State of the Arts: Table VI shows the state-of-the-art methods on the test set of these five data sets. We mainly provide the comparison methods of the following two types: one is the method that based on the state-of-the-art attention mechanism [1], [9], [13], [15], [18], [19], [43], [44] and the other one is the data-centric method [31], [32], [45]–[49], which achieves the state-of-the-art performance on these data sets. It is worth noting that most of the methods in the first type are designed for the natural scene images rather than the RSS data. Therefore, in order to compare with these methods fairly, we reimplemented them on the corresponding RSS data set and the results are marked them with "†" in Table VI. In terms of the experimental settings, except for the specific data set, the other settings are consistent with their article. The results of the second type of methods are obtained from their article. Experimental results in Table VI show that our proposed RTANet can achieve the new state-of-the-art performance on these data sets. In particular, RTANet-101 can significantly improve the performance, i.e., 3.16% on RSSCN7, 1.55% on UCML, 0.54% on WHU-RS19, 4.1% on AID, and 1.7% on NWPU-45 higher than baseline-2.

B. Experiments on Instance-Level RSS Object Detection

RSS object detection is an elaborate recognition task, which aims to assign a bounding box for each object (i.e., for "where is it") in the input image and predict each bounding box a class label (i.e., for "what is it").

1) Data Sets: Two benchmarks are used for RSS object detection. DOTA [26] contains a total of 2806 images

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:23:37 UTC from IEEE Xplore. Restrictions apply.



Fig. 5. Samples of the experimental data set for RSS object detection. Images are from the NWPU VHR-10 data set.

(1411 for the training set, 458 for the validation set, and 937 for the test set) in 15 categories, each of which has the resolution from 800 × 800 to 4000 × 4000. Since there is no ground truth publicly available for the test set of DOTA, in this article, we only report the detection results on the validation set. NWPU VHR-10 [27] contains a total of 800 images in ten categories, each of which has the average resolution of 1000 × 1000. Some samples of this data set are shown in Fig. 5. In terms of NWPU VHR-10, we randomly divide it into 80% for training (640 images) and 20% for test (160 images).

2) Evaluation Metrics: As in [39], we use the mAP over all categories as the prime evaluation metric. Following [50], the threshold intersection over union (IoU) is set to 0.5, i.e., if the IoU between the prediction and ground truth is larger than 0.5, this prediction will be considered as the true positive; otherwise, it will be considered as the false positive. Besides, model params, FLOPs, and inference time are also taken into consideration.

3) Backbone Networks: ResNet-50 [3] is selected as the primary backbone network. Besides, we also implement the RTA module on ResNet-101 [3] for the comparison with the state-of-the-art methods. As in RSS classification, all backbone networks are pretrained on ImageNet [41] in RSS object detection. In particular, to make the detection model can better recognize small objects in the image, following [51], convolutional layers of the last stage for the backbone network are replaced by dilated convolutions [52] with a common stride of 1 such that the backbone network can return a feature map of stride 16, i.e., the output feature map size of the backbone network is $1/16$ of the input.

4) Baseline: Faster R-CNN [51] with the dilated ResNet-50 as the backbone is used as the baseline (termed "baseline-3" in this article), where the region proposal network shares the same backbone as the Faster R-CNN system. Following [3], the implementation details of the head network are implemented as the same in its article.

5) Implementation Details: As in [39], the input images in the training stage are resized into 800 × 800. Following [50], all backbone networks are first pretrained on the ImageNet [41] and then fine-turned with a specific RSS data set. The proposed RTA module with $l = 2$ is added in the fourth stage of the backbone. The mix-pooling function is used as the sampling method, and the downsampling rate r is set to 2. The whole detection network is trained in an end-to-end manner on four NVIDIA Tesla P100 GPUs with the SyncBN [40]. Mini-batch is set to 2, and RoIs for each image are set to 512. We use a weight decay of 0.0001 and a momentum of 0.9. The learning rate is 0.01 for the first 60k mini-batches and 0.002 for the remainder 20k. Besides, as in [50], a similar data augmentation strategy is used, i.e., image rotation and flip. Furthermore, the warm-up heuristic [42], the zero γ, and the multiscale training strategy are also used in this section.

TABLE VII
COMPARISONS WITH STATE-OF-THE-ART METHODS ON THE VALIDATION SET OF DOTA [26]

methods	mAP	FLOPs	params (M)	time
baseline-3 [51]	0.3639	**152.3** G	**42.5** M	099 ms
FPN [39]	0.4691	279.4 G	44.4 M	112 ms
DeformNet [53]	0.6694	281.4 G	48.5 M	097 ms
Azimi *et al.* [54]	0.7054	279.1 G	48.9 M	120 ms
DM-FPN [50]	0.7927	280.3 G	44.7 M	145 ms
RTANet-50 (ours)	0.7527	153.5 G	43.3 M	154 ms
RTANet-101 (ours)	**0.7906**	188.2 G	65.0 M	**095** ms

6) Comparisons With State of the Arts: Tables VII and VIII show the experimental results of object detection on DOTA and NWPU VHR-10, respectively. Compared with state-of-the-art methods [6], [39], [50], [53]–[57], we can observe that RTANet-50 can achieve very competitive results, i.e., 0.7527 mAP on the validation set of DOTA and 0.8261 mAP on the test set of NWPU VHR-10. They correspond to the model performance improvements by 0.3888 and 0.3369 mAP when compared to baseline-3. Besides, RTANet-101 achieves the state-of-the-art performance on the test set of NWPU VHR-10, i.e., 0.8420 mAP. As for DOTA, RTANet-101 outperforms the majority of the state-of-the-art methods [39], [53], [54]. However, its performance is inferior

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:23:37 UTC from IEEE Xplore. Restrictions apply.

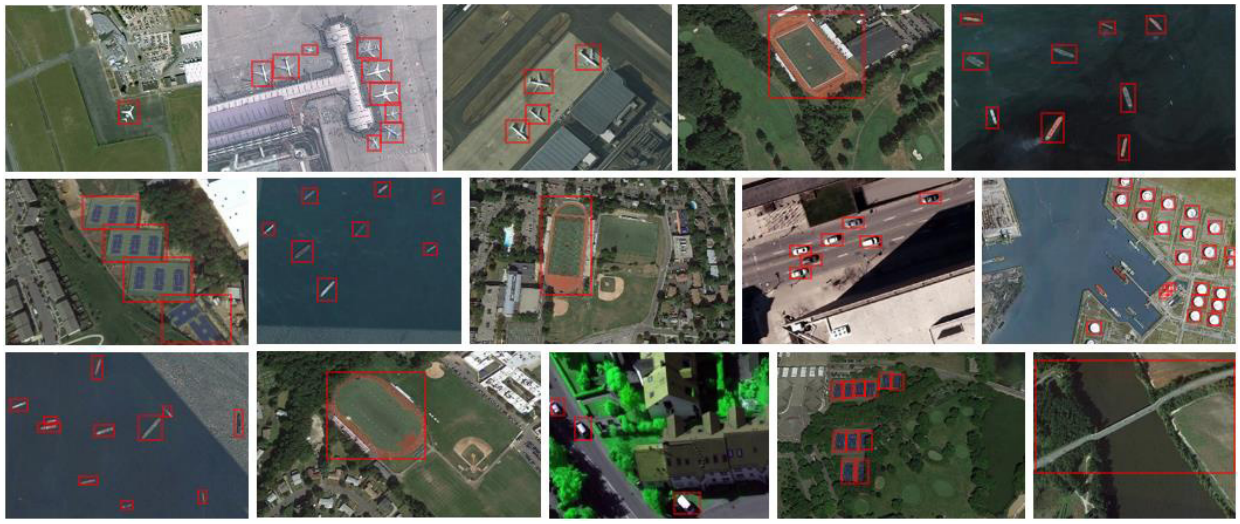


Fig. 6. Qualitative results of RTANet on RSS object detection. Images are from the NWPU VHR-10 data set. Since the predicted class labels and scores for these objects are too small to see, we skip it in the visualization.

TABLE VIII
COMPARISONS WITH STATE-OF-THE-ART METHODS ON THE TEST SET OF NWPU VHR-10 [27]

methods	mAP	FLOPs	params (M)	time
baseline-3 [51]	0.4892	**152.3** G	**42.5** M	105 ms
RICNN [55]	0.5982	162.7 G	44.6 M	112 ms
Long *et al.* [6]	0.7286	167.5 G	46.9 M	101 ms
Li *et al.* [56]	0.8271	175.0 G	49.2 M	099 ms
MS-VANs [57]	0.8378	173.2 G	48.8 M	095 ms
RTANet-50 (ours)	0.8261	152.3 G	42.6 M	085 ms
RTANet-101 (ours)	**0.8420**	163.8 G	55.8 M	**071** ms

to that of [50]. One possible reason is that double multi-scale (DM)-FPN in [50] is based on the feature pyramid network [39], which is a more powerful feature representation method. Therefore, it is common that DM-FPN can achieve a better detection result.

More importantly, in terms of computational complexity, RTANet is far computationally cheap than other methods. As shown in Table VII, compared to baseline-3, RTANet-50 only increases the parameters by 1.88% and the FLOPs by 0.79%. From Table VIII, RTANet-50 only increases 0.1 M params, and no model FLOPs improve when compared to baseline-3. Experimental results on RTANet-101 can also confirm the effectiveness and efficiency of RTA. Therefore, the superiority of RTANet is further verified. Besides, the experimental results on the inference time show that RTANet has not only higher efficiency but also shorter inference time when compared to other models. The qualitative results are shown in Fig. 6. As we expected, RTANet can predict extremely high-quality results, especially for some extremely large (e.g., basketball court and ground track field) and small objects (e.g., airplane, ship, baseball diamond, and vehicle).

V. CONCLUSION

We proposed a simple yet effective global context acquisition module, named RTA, which can be used to capture the long-range dependence in a more efficient manner. Based on RTA, we further proposed an efficient RTANet for RSS recognition. Extensive experimental results on the image-level RSS classification and the instance-level RSS object detection demonstrated its superiority.

RTA is a general context acquisition module, which can be inserted into any hierarchy of a deep CNN model for arbitrary visual recognition task. Therefore, in the future, we will consider applying the RTA module to other RSS recognition tasks, e.g., the change detection and RSS segmentation. Moreover, RTA can also be used in the multispectral RSS image processing task. Although the RTA module is extraordinarily superior in effectiveness and efficiency, no business is perfect. Therefore, it is also a promising direction to continue exploring more superior context acquisition methods in the future.

REFERENCES

[1] D. Zhang, N. Li, and Q. Ye, "Positional context aggregation network for remote sensing scene classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 6, pp. 943–947, Jun. 2020.
[2] B. Du, S. Cai, C. Wu, L. Zhang, and D. Tao, "Object tracking in satellite videos based on a multiframe optical flow tracker," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 8, pp. 3043–3055, Aug. 2019.
[3] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
[4] L. Mou, P. Ghamisi, and X. X. Zhu, "Deep recurrent neural networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3639–3655, Jul. 2017.
[5] Y. Xu, B. Du, and L. Zhang, "Beyond the patchwise classification: Spectral-spatial fully convolutional networks for hyperspectral image classification," *IEEE Trans. Big Data*, vol. 6, no. 3, pp. 492–506, Sep. 2020.
[6] Y. Long, Y. Gong, Z. Xiao, and Q. Liu, "Accurate object localization in remote sensing images based on convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 5, pp. 2486–2498, May 2017.
[7] L. Mou, L. Bruzzone, and X. X. Zhu, "Learning spectral-spatial-temporal features via a recurrent convolutional neural network for change detection in multispectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 924–935, Feb. 2019.
[8] B. Du, L. Ru, C. Wu, and L. Zhang, "Unsupervised deep slow feature analysis for change detection in multi-temporal remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 12, pp. 9976–9992, Dec. 2019.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:23:37 UTC from IEEE Xplore. Restrictions apply.

[9] I. Bello, B. Zoph, Q. Le, A. Vaswani, and J. Shlens, "Attention augmented convolutional networks," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct/Nov. 2019, pp. 3285–3294.

[10] F. Yu, V. Koltun, and T. Funkhouser, "Dilated residual networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 472–480.

[11] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "DeepLab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected CRFs," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 4, pp. 834–848, Apr. 2018.

[12] C. Peng, X. Zhang, G. Yu, G. Luo, and J. Sun, "Large kernel matters—Improve semantic segmentation by global convolutional network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 4353–4361.

[13] X. Wang, R. Girshick, A. Gupta, and K. He, "Non-local neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7794–7803.

[14] A. Vaswani *et al.*, "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, 2017, pp. 5998–6008.

[15] Y. Cao, J. Xu, S. Lin, F. Wei, and H. Hu, "GCNet: Non-local networks meet squeeze-excitation networks and beyond," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshop (ICCVW)*, Oct. 2019, pp. 1971–1980.

[16] J. Fu *et al.*, "Dual attention network for scene segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 3146–3154.

[17] K. Yue, M. Sun, Y. Yuan, F. Zhou, E. Ding, and F. Xu, "Compact generalized non-local network," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, 2018, pp. 6510–6519.

[18] Z. Huang, X. Wang, L. Huang, C. Huang, Y. Wei, and W. Liu, "CCNet: Criss-cross attention for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 603–612.

[19] Z. Zhu, M. Xu, S. Bai, T. Huang, and X. Bai, "Asymmetric non-local neural networks for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 593–602.

[20] A. Paszke *et al.*, "Pytorch: An imperative style, high-performance deep learning library," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, 2019, pp. 8026–8037.

[21] Q. Zou, L. Ni, T. Zhang, and Q. Wang, "Deep learning based feature selection for remote sensing scene classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 11, pp. 2321–2325, Nov. 2015.

[22] Y. Yang and S. Newsam, "Bag-of-visual-words and spatial extensions for land-use classification," in *Proc. 18th SIGSPATIAL Int. Conf. Adv. Geographic Inf. Syst. (GIS)*, 2010, pp. 270–279.

[23] G. Sheng, W. Yang, T. Xu, and H. Sun, "High-resolution satellite scene classification using a sparse coding based multiple feature combination," *Int. J. Remote Sens.*, vol. 33, no. 8, pp. 2395–2412, Apr. 2012.

[24] G.-S. Xia *et al.*, "AID: A benchmark data set for performance evaluation of aerial scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3965–3981, Jul. 2017.

[25] G. Cheng, J. Han, and X. Lu, "Remote sensing image scene classification: Benchmark and state of the art," *Proc. IEEE*, vol. 105, no. 10, pp. 1865–1883, Oct. 2017.

[26] G.-S. Xia *et al.*, "DOTA: A large-scale dataset for object detection in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3974–3983.

[27] G. Cheng, J. Han, P. Zhou, and L. Guo, "Multi-class geospatial object detection and geographic image classification based on collection of part detectors," *ISPRS J. Photogramm. Remote Sens.*, vol. 98, pp. 119–132, Dec. 2014.

[28] H. Zhou, Y. Yuan, and C. Shi, "Object tracking using SIFT features and mean shift," *Comput. Vis. Image Understand.*, vol. 113, no. 3, pp. 345–352, Mar. 2009.

[29] Y.-G. Jiang, C.-W. Ngo, and J. Yang, "Towards optimal bag-of-features for object categorization and semantic video retrieval," in *Proc. 6th ACM Int. Conf. Image Video Retr. (CIVR)*, 2007, pp. 494–501.

[30] J. Y.-H. Ng, F. Yang, and L. S. Davis, "Exploiting local features from deep networks for image retrieval," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2015, pp. 53–61.

[31] Y. Boualleg, M. Farah, and I. R. Farah, "Remote sensing scene classification using convolutional features and deep forest classifier," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 12, pp. 1944–1948, Dec. 2019.

[32] D. Zeng, S. Chen, B. Chen, and S. Li, "Improving remote sensing scene classification by integrating global-context and local-object features," *Remote Sens.*, vol. 10, no. 5, p. 734, 2018.

[33] M. D. Zeiler and R. Fergus, "Visualizing and understanding convolutional networks," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2014, pp. 818–833.

[34] J. Hu, L. Shen, and G. Sun, "Squeeze- and-excitation networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7132–7141.

[35] W. Zhang, D. Zhang, and X. Xiang, "Cascaded and dual: Discrimination oriented network for brain tumor classification," in *Proc. 11th Asian Conf. Mach. Learn. (ACML)*, 2019, pp. 363–378.

[36] P. Sauseng, W. Klimesch, W. R. Gruber, and N. Birbaumer, "Cross-frequency phase synchronization: A brain mechanism of memory matching and attention," *NeuroImage*, vol. 40, no. 1, pp. 308–317, Mar. 2008.

[37] Z. Wu, C. Shen, and A. van den Hengel, "Wider or deeper: Revisiting the ResNet model for visual recognition," *Pattern Recognit.*, vol. 90, pp. 119–133, Jun. 2019.

[38] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 4700–4708.

[39] T.-Y. Lin, P. Dollar, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2117–2125.

[40] H. Zhang *et al.*, "Context encoding for semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7151–7160.

[41] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "ImageNet: A large-scale hierarchical image database," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2009, pp. 248–255.

[42] T. He, Z. Zhang, H. Zhang, Z. Zhang, J. Xie, and M. Li, "Bag of tricks for image classification with convolutional neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 558–567.

[43] R. Liu *et al.*, "An intriguing failing of convolutional neural networks and the coordconv solution," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, 2018, pp. 9605–9616.

[44] N. Parmar *et al.*, "Image transformer," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2018, pp. 1–10.

[45] G. Cheng, C. Yang, X. Yao, L. Guo, and J. Han, "When deep learning meets metric learning: Remote sensing image scene classification via learning discriminative CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 5, pp. 2811–2821, May 2018.

[46] D. Ma, P. Tang, and L. Zhao, "SiftingGAN: Generating and sifting labeled samples to improve the remote sensing image scene classification baseline *in vitro*," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 7, pp. 1046–1050, Jul. 2019.

[47] W. Zhang, P. Tang, and L. Zhao, "Remote sensing image scene classification using CNN-CapsNet," *Remote Sens.*, vol. 11, no. 5, p. 494, Feb. 2019.

[48] X. Lu, W. Ji, X. Li, and X. Zheng, "Bidirectional adaptive feature fusion for remote sensing scene classification," *Neurocomputing*, vol. 328, pp. 135–146, Feb. 2019.

[49] Y. Yu and F. Liu, "A two-stream deep fusion framework for high-resolution aerial scene classification," *Comput. Intell. Neurosci.*, vol. 2018, pp. 1–13, Jul. 2018.

[50] X. Zhang *et al.*, "Geospatial object detection on high resolution remote sensing imagery based on double multi-scale feature pyramid network," *Remote Sens.*, vol. 11, no. 7, p. 755, Mar. 2019.

[51] R. Girshick, "Fast R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1440–1448.

[52] F. Yu and V. Koltun, "Multi-scale context aggregation by dilated convolutions," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2016, pp. 1–13.

[53] J. Dai *et al.*, "Deformable convolutional networks," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 764–773.

[54] S. M. Azimi, E. Vig, R. Bahmanyar, M. Körner, and P. Reinartz, "Towards multi-class object detection in unconstrained remote sensing imagery," in *Proc. Asian Conf. Comput. Vis. (ACCV)*, 2018, pp. 150–165.

[55] G. Cheng, P. Zhou, and J. Han, "Learning rotation-invariant convolutional neural networks for object detection in VHR optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 12, pp. 7405–7415, Dec. 2016.

[56] X. Li and S. Wang, "Object detection using convolutional neural networks in a coarse-to-fine manner," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 11, pp. 2037–2041, Nov. 2017.

[57] C. Wang, X. Bai, S. Wang, J. Zhou, and P. Ren, "Multiscale visual attention networks for object detection in VHR remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 2, pp. 310–314, Feb. 2019.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:23:37 UTC from IEEE Xplore. Restrictions apply.



Liyong Fu received the B.S. degree in forestry from Shanxi Agricultural University, Jinzhong, China, in 2007, the M.S. degree in forest biometrics from Nanjing Forestry University, Nanjing, China, in 2009, and the Ph.D. degree in forest biometrics from the Chinese Academy of Forestry, Beijing, China, in 2012.

He is a Full Professor of forest biometrics with the Department of Forest Management and Statistics, Chinese Academy of Forestry. He has authored or coauthored a total of 43 SCI articles in prestigious peer-reviewed international journals including *Briefings in Bioinformatics*, *Neural Networks*, IEEE TRANSACTIONS ON IMAGE PROCESSING, IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, and *Frontiers in Plant Science* during the recent five years.



Dong Zhang received the B.S. and M.S. degrees in computer science and technology from Nanjing Forestry University, Nanjing, China. He is pursuing the Ph.D. degree with the School of Computer Science and Engineering, Nanjing University of Science and Technology, Nanjing, China.

From September 2018 to September 2020, he was supported by China Scholarship Council as a joint Ph.D. student with Nanyang Technological University, Singapore. His research interests include machine learning and computer vision, especially in network architecture design, object detection, and semantic segmentation. He has published several journal articles and conference papers including IEEE TRANSACTIONS ON CYBERNETICS, *Computers and Electronics in Agriculture*, Association for the Advancement of Artificial Intelligence (AAAI), European Conference on Computer Vision (ECCV), and Conference on Neural Information Processing Systems (NeurIPS).



Qiaolin Ye (Member, IEEE) received the Ph.D. degree in pattern recognition and intelligence system from the Nanjing University of Science and Technology, Nanjing, China, in 2013.

He is a Full Professor with the Computer Science Department, Nanjing Forestry University, Nanjing, China. He has authored more than 50 scientific articles. His research interests include machine learning, data mining, and pattern recognition. Some of them are published in the IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, IEEE TRANSACTIONS ON INFORMATION FORENSICS AND SECURITY, and IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:23:37 UTC from IEEE Xplore. Restrictions apply.